EWB CLOUD LABS

AWS VPC Peering - Student Project Report

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Batch	Batch 1
AWS Region	Mumbai ap-south-1
Completion date	29/10/2026
Project	EWB-VPC-Peering

1. Project Objective

Build private connectivity between an application server in VPC A and a workload in VPC B using VPC Peering. Keep both application servers private and administer them through a public bastion host in VPC A. Demonstrate baseline failure, successful two-way connectivity, SSH access and controlled ICMP failure/recovery.

2. Architecture and Address Plan

Laptop -> SSH (public IPv4) -> Bastion (10.10.1.10) -> SSH ProxyJump to App-A (10.10.2.10) or App-B (10.20.1.10). App-A <-> VPC Peering <-> App-B uses private IPv4 traffic. VPC B has no Internet Gateway.

Resource	CIDR / IP	Actual AWS ID
ewb-vpc-a	10.10.0.0/16	vpc-00e76f1359ffaa8af
ewb-a-public	10.10.1.0/24	subnet-0900864c770989e98
ewb-a-private	10.10.2.0/24	subnet-0ceb911a9a1a9e44b
ewb-vpc-b	10.20.0.0/16	vpc-086d032a3b93a021c
ewb-b-private	10.20.1.0/24	subnet-03ead42b4648ba70d
ewb-bastion	10.10.1.10 + [Public IPv4]	i-0f4933b95cac8ca9b
ewb-app-a	10.10.2.10, no public IP	i-07b3b44332272e3ad
ewb-app-b	10.20.1.10, no public IP	i-06322f17b1d3d5d41

Record actual IPs if AWS assigned different addresses. Update all /32 rules and commands accordingly.

3. Network and Route Tables

Internet Gateway: ewb-igw-a [igw-034f5aabd7c9293c3], attached to VPC A. Explicit subnet associations are required. No NAT Gateway is used.

Route table / associated subnet	Destination	Target
ewb-rt-a-public / ewb-a-public	10.10.0.0/16	local
	0.0.0.0/0	ewb-igw-a [igw-034f5aabd7c9293c3]
	10.20.0.0/16	ewb-a-b-peer [pcx-02db178e624e37c47]
ewb-rt-a-private / ewb-a-private	10.10.0.0/16	local
	10.20.0.0/16	[pcx-02db178e624e37c47]
ewb-rt-b-private / ewb-b-private	10.20.0.0/16	local
	10.10.0.0/16	[pcx-02db178e624e37c47]

4. Security Groups

Security Group	Inbound type	Source
ewb-sg-bastion	SSH TCP 22	0.0.0.0/32
ewb-sg-app-a	SSH TCP 22	10.10.1.10/32
ewb-sg-app-a	ICMP IPv4 Echo request	10.20.1.10/32
ewb-sg-app-b	SSH TCP 22	10.10.1.10/32
ewb-sg-app-b	ICMP IPv4 Echo request	10.10.2.10/32

Default outbound allow-all rules and default allow-all network ACLs are retained for this lab. Security Groups are stateful.

5. EC2 and SSH Configuration

Three Amazon Linux 2023 x86 EC2 instances use the same ewb-lab RSA key pair. Bastion is in the public subnet with public IPv4 enabled. App-A and App-B are in private subnets with public IPv4 disabled. Record the actual instance type, instance IDs and Bastion public IP. Never include the PEM private key in this report.

Administration path: local MobaXterm SSH session to Bastion using ec2-user and the local PEM file; App-A/App-B sessions use Bastion as SSH gateway/jump host, with the local key.

6. VPC Peering

Peering connection ewb-a-b-peer: requester ewb-vpc-a; accepter ewb-vpc-b; same AWS account and Region. Actual connection ID:pcx-02db178e624e37c47. Peering must be accepted and both forward and return routes configured.

7. Validation Results - ENTER ACTUAL OBSERVATIONS

Test / where	Command or action	Actual result / evidence
Baseline / App-A before peering	ping -c 4 10.20.1.10	
Forward / App-A after routes	ping -c 4 10.20.1.10	
Reverse / App-B after routes	ping -c 4 10.10.2.10	
Private SSH / laptop	MobaXterm session to App-B via Bastion	
Isolation / EC2 details	Inspect App-A and App-B Public IPv4	
Controlled failure	Remove App-B inbound ICMP Echo request from App-A; repeat forward ping	
Recovery	Restore same ICMP rule; repeat forward ping	

8. Explanation of Traffic Paths

Forward path: App-A subnet route for 10.20.0.0/16 sends traffic to the active peering connection; App-B Security Group permits ICMP Echo requests from App-A. Return path: VPC B route for 10.10.0.0/16 sends responses back over peering. The Bastion public subnet also requires a 10.20.0.0/16 peering route to SSH into App-B through the jump host. Peering does not provide VPC B internet access through VPC A.

9. Troubleshooting / Learning Outcomes

MY laptop was not connecting to instances, As guided my instructor i used moboxtream to connect to them

[Write a short reflection on CIDR planning, route associations, stateful security groups, bastion access and bidirectional routing.]

10. Cleanup Evidence

After saving evidence: terminate three lab instances; confirm lab EBS volumes/snapshots and public IPv4 allocations are cleared; remove peer routes and peering; remove lab Security Groups and subnets/route tables; detach/delete IGW; delete both VPCs; check Billing and project tags. Record actual cleanup date.